

Time-independent Schrodinger equation

Consider the Schrodinger equation

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = H\psi(x, t)$$

Let $\psi(x, t)$ be separable as

$$\psi(x, t) = \varphi(t)\phi(x)$$

Let it be the case that for the time derivative we have

$$i\hbar \frac{\partial}{\partial t} \varphi(t) = E\varphi(t)$$

where E is a constant. Then

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = E\varphi(t)\phi(x) = E\psi(x, t)$$

Hence by the Schrodinger equation

$$E\psi(x, t) = H\psi(x, t)$$

If H is independent of t then $\varphi(t)$ cancels and we have the time-independent Schrodinger equation

$$E\phi(x) = H\phi(x)$$

In summary, let $\psi(x, t)$ be a wavefunction such that

1. $\psi(x, t)$ solves the Schrodinger equation
2. $\psi(x, t)$ is separable as $\varphi(t)\phi(x)$
3. $i\hbar \partial_t \varphi(t) = E\varphi(t)$ where E is a constant
4. H is independent of t

Then $\phi(x)$ solves the time-independent Schrodinger equation

$$E\phi(x) = H\phi(x)$$

For example, the Hamiltonian for the harmonic oscillator is

$$H = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{1}{2}m\omega^2 x^2$$

The wavefunctions are

$$\psi_n(x, t) = C_n \exp\left(-\frac{iE_n t}{\hbar}\right) \exp\left(\frac{m\omega x^2}{2\hbar}\right) \frac{d^n}{dx^n} \exp\left(-\frac{m\omega x^2}{\hbar}\right)$$

where C_n is a normalization constant and

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right)$$

Taking the time derivative of $\psi_n(x, t)$ we find

$$i\hbar \frac{\partial}{\partial t} \psi_n(x, t) = E_n \psi_n(x, t)$$

Hence

$$E_n \psi_n(x, t) = H \psi_n(x, t)$$

Since H is independent of t we have

$$E_n \phi_n(x) = H \phi_n(x)$$

where

$$\phi_n(x) = C_n \exp\left(\frac{m\omega x^2}{2\hbar}\right) \frac{d^n}{dx^n} \exp\left(-\frac{m\omega x^2}{\hbar}\right)$$

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